

Name: Anaya Stahl (please print)

ECE 2202 – HW 6

Chapter 7: Problems 7.44, 7.48, 7.62, 7.65, 7.68, 7.80, 7.83

****For all plots, use Matlab or other graphing software****

***Note:** These are problems from the Textbook PDF which you can download here if you have not already: <https://cad2e.eecs.umich.edu/>

*7.44 In the circuit of Fig. P7.44, what should the value of L be at $\omega = 10^4$ rad/s so that $i(t)$ is in-phase with $v_s(t)$?

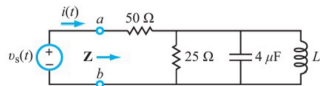


Figure P7.44 Circuit for Problem 7.44.

$$Z_c + Z_L$$

$$Z_c = \frac{j}{\omega(4\mu F)} \rightarrow -j25[\Omega]$$

$$Z_c \parallel Z_L \rightarrow j25[\Omega] \cdot (j10^4 L) = -j25 \cdot 0.002[\Omega] \cdot L$$

$$Z_L = j\omega L \rightarrow j(10^4)L$$

$$Z_{R1} \parallel Z_{R2}$$

$$R_{eq} = \frac{50[\Omega] \cdot 25[\Omega]}{50[\Omega] + 25[\Omega]} = 16.67[\Omega]$$

$$Z = \frac{25(j\omega L)}{(j\omega L) + 25 - 25(\omega^2 LC)} \rightarrow 50[\Omega] + \left(\frac{25[\Omega](j\omega L(25 - 25\omega^2 LC) - j\omega L)}{(25 - 25\omega^2 LC)^2 - \omega^2 L^2} \right)$$

$$Z_c \parallel R_{eq} = \frac{-j25.0027[\Omega] \cdot 16.67[\Omega]}{16.67[\Omega] - j25.0027[\Omega]} = 11.56 - j7.69[\Omega]$$

$$Z_T = \left(50[\Omega] + \frac{25\omega^2 L^2}{(25 - 25\omega^2 LC)^2 - \omega^2 L^2} \right) + \left(\frac{j16.25\omega L - 625\omega^3 L^2 C}{(25 - 25\omega^2 LC)^2 - \omega^2 L^2} \right) = 0$$

it in phase the imaginary part = 0

$$\frac{16.25\omega L - 625\omega^3 L^2 C}{(25 - 25\omega^2 LC)^2 - \omega^2 L^2} = 0$$

$$(16.25\omega L - 625\omega^3 L^2 C) = 0$$

$$16.25\omega L(1 - \omega^2 LC) = 0$$

$$\omega^2 LC = 1 \rightarrow L = \frac{1}{\omega^2 C}$$

$$L = \frac{1}{(10^4)^2 (4\mu F)} \rightarrow 2.5[mH]$$

7.48 Determine the Thévenin equivalent of the circuit in Fig. P7.48 at terminals (a,b), given that

$$v_s(t) = 12 \cos 2500t \text{ V}$$

$$i_s(t) = 0.5 \cos(2500t - 30^\circ) \text{ A}$$

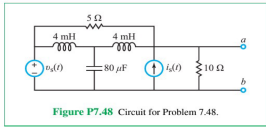


Figure P7.48 Circuit for Problem 7.48.

$$Z_{L1} = j\omega L = j(2500)(4 \text{ mH}) = j10 \text{ } \Omega$$

$$Z_{L2} = j\omega L = j(2500)(4 \text{ mH}) = j10 \text{ } \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j2500(80 \mu\text{F})} \rightarrow -j5 \text{ } \Omega$$

7.48 Determine the Thévenin equivalent of the circuit in Fig. P7.48 at terminals (a,b), given that

$$v_s(t) = 12 \cos 2500t \text{ V}$$

$$i_s(t) = 0.5 \cos(2500t - 30^\circ) \text{ A}$$

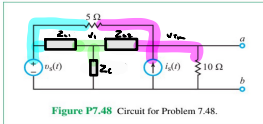


Figure P7.48 Circuit for Problem 7.48.

$$\frac{V_1}{Z_C} + \frac{V_1 - V_S}{Z_{L1}} + \frac{V_1 - V_{Th}}{Z_{L2}} = 0$$

$$\frac{V_1}{-j5 \text{ } \Omega} + \frac{V_1 - 12 \text{ V}}{j10 \text{ } \Omega} + \frac{V_1 - V_{Th}}{j10 \text{ } \Omega} = 0$$

$$\frac{V_{Th} - V_1}{j10 \text{ } \Omega} + \frac{V_{Th} - 12 \text{ V}}{5 \text{ } \Omega} + \frac{V_{Th}}{10 \text{ } \Omega} = 0$$

$$V_1 = (-12 - j60) \text{ V}$$

$$V_{Th} = -12 \text{ V}$$

$$V_{Th} = 12 \text{ V}$$

7.48 Determine the Thévenin equivalent of the circuit in Fig. P7.48 at terminals (a,b), given that

$$v_s(t) = 12 \cos 2500t \text{ V}$$

$$i_s(t) = 0.5 \cos(2500t - 30^\circ) \text{ A}$$

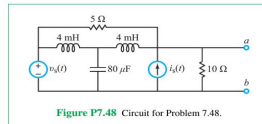


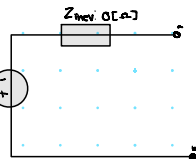
Figure P7.48 Circuit for Problem 7.48.

Thévenin:

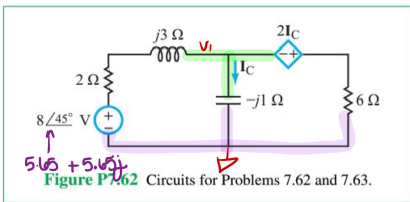
$$Z_{Th} = Z_{L1} \parallel Z_C \rightarrow \left(\frac{j10 \text{ } \Omega \cdot -j5 \text{ } \Omega}{j10 \text{ } \Omega - j5 \text{ } \Omega} \right) = -j10 \text{ } \Omega$$

$$= \left(\frac{1}{j5 \text{ } \Omega} + \frac{1}{10 \text{ } \Omega} + \frac{1}{j10 \text{ } \Omega - j10 \text{ } \Omega} \right)^{-1}$$

$$Z_{Th} = 0 \text{ } \Omega$$



7.62 Apply nodal analysis to determine I_C in the circuit of Fig. P7.62.



$$I_C = \frac{1.356 + j1.356}{-j1[\Omega]} = -1.356 + j1.356[A]$$

$$\approx 1.917 \angle 135^\circ$$

$$I_C = \frac{V_1}{-j1[\Omega]}$$

$$\left(\frac{V_1 - V_S}{j3[\Omega]} + \frac{V_1}{-j1[\Omega]} - 2I_C + \frac{V_1}{6[\Omega]} = 0 \right)$$

$$6V_1 - (33.9 + j33.9)[V] + 6V_1 - 12\left(\frac{V_1}{-j1[\Omega]}\right) + V_1 = 0$$

$$6V_1 - (33.9 + j33.9)[V] + 6V_1 + 12V_1 + V_1 = 0$$

$$\frac{25V_1 = 33.9 + j33.9}{25} \rightarrow V_1 = 1.356 + j1.356[V]$$

7.65 Use mesh analysis to obtain an expression for the phasor V_{out} in the circuit of Fig. P7.65, in terms of V_s and R , given that $R = \omega L = 1/\omega C$.

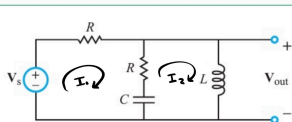


Figure P7.65 Circuit for Problem 7.65.

$$-V_s + I_1 R + (I_1 - I_2) \left[R + \frac{1}{j\omega C} \right] = 0 \quad \omega = \frac{1}{RC}$$

$$-V_s + I_1 R + (I_1 - I_2) [R - jR] = 0$$

$$-V_s + I_1 R + I_1 R - I_2 R - jR(I_1 - I_2) = 0$$

$$\textcircled{1} \quad I_1 (-j + 2) - I_2 (-j + 1) = \frac{V_s}{R}$$

$$\textcircled{2} \quad (I_2 - I_1) \left(R + \frac{1}{j\omega C} \right) + I_2 (j\omega L) = 0$$

$$\frac{1}{j\omega C} = \omega L = R$$

$$I_2 R - I_2 jR - I_1 R + I_1 jR + I_2 jR = 0$$

$$I_2 R = I_1 (R - jR) \rightarrow I_2 = I_1 (-j + 1) = I_1 \frac{1-j}{1-j}$$

$$I_2 \left[\frac{2-j}{1-j} - (1-j) \right] = \frac{V_s}{R}$$

$$I_2 \left(\frac{(2-j) - (1-j)^2}{1-j} - j \right) = \frac{V_s}{R}$$

$$I_2 = \frac{V_s}{R} \left(\frac{1-j}{2-j} \right) \cdot \frac{(2-j)}{2-j} \rightarrow \frac{V_s}{R} \left(\frac{2-2j-j+j^2}{4-j^2} \right)$$

$$= \frac{V_s}{R} \left(\frac{2-3j-1}{4+1} \right) \rightarrow \frac{V_s}{R} \left(\frac{1-3j}{5} \right)$$

$$V_{out} = I_2 (j\omega L)$$

$$V_{out} = \frac{V_s}{R} \left(\frac{1-3j}{5} \right) (j\omega L) = \frac{V_s \omega L}{5R} (3-j)$$

$$j^2 = -1$$

$$= 0.63V_s \angle 18.14^\circ$$

*7.68 Determine $i_x(t)$ in the circuit of Fig. P7.68, given that $v_s(t) = 6\cos 5 \times 10^3 t$ V.

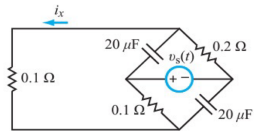
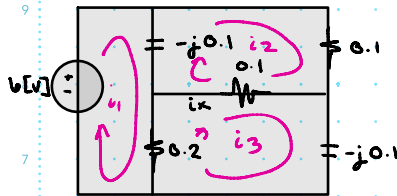


Figure P7.68 Circuit for Problem 7.68.

$$\omega = 5000 [\text{rad/s}]$$

$$Z_C = \frac{j}{\omega C} \rightarrow \frac{j}{5000(20 \times 10^{-6})} = -j0.1$$



KVL

$$\textcircled{1} \rightarrow -j0.1(I_1 - I_2) + 0.2(I_1 - I_3) = 6 \angle 0^\circ$$

$$I_1(0.2 - 0.1j) + 0.1jI_2 - 0.2I_3 = 6 \angle 0^\circ$$

$$\textcircled{2} \rightarrow 0.1jI_1 - 0.1I_2 + I_3(0.3 - 0.1j) = 0$$

$$I_1: 52.77 \angle 54.6^\circ \text{ [A]}$$

$$I_2: 26.06 \angle 34.4^\circ \text{ [A]}$$

$$I_3: 41.2 \angle 69.1^\circ \text{ [A]}$$

$$I_x = I_3 - I_2 = 24.72 \angle 106^\circ \text{ [A]}$$

*7.80 The input signal in the op-amp circuit of Fig. P7.80 is given by

$$v_{in}(t) = V_0 \cos \omega t.$$

Assuming the op amp is operating within its linear range, obtain an expression for $v_{out}(t)$ by applying the phasor-domain technique and then evaluate it for $\omega RC = 1$.

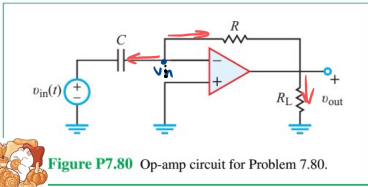


Figure P7.80 Op-amp circuit for Problem 7.80.

$$V_2 = 0 \rightarrow V_1 = 0$$

$$\frac{V_1 - V_{in}}{\frac{1}{j\omega C}} + \frac{V_1 - V_{out}(t)}{R} = 0$$

$$\frac{V_{out}(t)}{V_{in}(t)} = -j\omega CR$$

$$V_{out}(t) = -j V_{in}(t)$$

$$= \angle -90^\circ$$

$$V_{out}(t) = V_0 \sin \omega t$$

* **7.83** For $v_i(t) = V_0 \cos \omega t$, obtain an expression for $v_{out}(t)$ in the circuit of **Fig. P7.83** and then evaluate it for $V_0 = 2$ V, $\omega = 377$ rad/s, $R_1 = 2$ k Ω , $R_2 = 10$ k Ω , and $C = 0.5$ μ F.

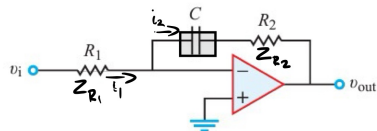


Figure P7.83 Circuit for Problem 7.83.

$$Z = R_2 - \frac{j}{\omega C} = 10e3 \left(\frac{-j}{j\pi(0.5e-6)} \right) = (10e3 - j5305) [\Omega]$$

$$I_1 = \frac{v_i}{R_1} \quad I_2 = \frac{0 - v_o}{Z} = \frac{-v_o}{Z} \quad I_1 = I_2$$

$$\frac{v_i}{R_1} = \frac{-v_o}{Z} \quad v_o = \frac{-Z}{R_1} \cdot v_i \rightarrow \frac{-(10e3 - j5305)}{20e3} \cdot v_i$$

$$v_o = (5 - j2.65) v_i = (5 - j2.65) (2 \cos 377t) = (10 - j5.3) \left[\frac{e^{j377t} + e^{-j377t}}{2} \right]$$

$$v_o = (5 - j2.65) \left[e^{j377t} + e^{-j377t} \right]$$